

TIMOTHY MARKFELD

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SUMMARY

Senior Software Engineer focused on distributed systems, backend platforms, and production operations. Experience designing and operating high-throughput services in healthcare and SaaS, including shipping integrations into HIPAA-regulated environments and EHR vendor platforms. Currently ship in C# and TypeScript on AWS.

EXPERIENCE

Senior Software Engineer

Apr. 2025 – Present

FirmPilot

Remote

- Built and owned fault-tolerant event-driven worker systems on AWS (SQS/SNS, ECS, RDS, Parameter Store), migrating selected ECS workloads to Kubernetes with retry logic and production observability.
- Designed GraphQL APIs and React/Next.js TypeScript clients with tight database access patterns, holding p99 query latency under 25ms in a multi-tenant product.
- Shipped LLM content pipelines (generation, fact-checking, media), and automated publishing.
- Led architecture cleanup and large-scale refactors, introducing CQRS patterns to improve long-term system maintainability.
- Merged 2,000+ PRs across backend, frontend, and infrastructure while conducting daily code reviews and mentoring.
- On-call for 45+ production incidents: 97.8% ack rate, 13s MTTA, 22min MTTR, zero timeout escalations.

Data Integration Engineer → Data Integration Engineer II

Sep. 2023 – Apr. 2025

HCA Healthcare

Nashville, TN

- Owned a HIPAA-regulated integration platform of 200+ .NET microservices processing 1M+ events/day over Kafka and IBM MQ.
- Shipped clinical data exchange into external EHR systems using FHIR, HL7, and IHE, working inside vendor networking and engineering constraints.
- Led migration of legacy integration services to Apache NiFi, including Terraform, Azure AD, and GCP, with Dynatrace alerting and automated failovers.
- Worked with vendor networking and engineering teams to get connectivity onto the new platform; mentored engineers on interoperability and architecture

Full Stack Engineer

May 2020 – Aug. 2023

Corolla Ice Delivery

Corolla, NC

- Owned a mobile app from design through deployment to the App Store and Google Play that cut delivery time by up to 3 hours/day.
- Replaced manual scheduling with route optimization used in daily operations.

EDUCATION

The University of Alabama

Tuscaloosa, AL

M.S. Management Information Systems, GPA: 4.0/4.0

Jan. 2021 – May 2023

B.S. Management Information Systems, GPA: 3.94/4.0

Sep. 2019 – May 2023

PROJECTS

HL7Kit / HL7Kit.Fhir | .NET, HL7, FHIR

github.com/tcmarkfeld/HL7Kit

- Strongly typed HL7 v2 parser with full segment coverage, dynamic delimiter handling, Z-segment support, and FHIR R4 conversion for .NET. Published on NuGet.

Conductor | .NET, Terraform, MassTransit

github.com/tcmarkfeld/Conductor

- CLI that generates least-privilege IAM policies and Terraform resources from MassTransit/SQS/SNS config for .NET teams deploying to AWS. Published on NuGet.

TECHNICAL SKILLS

Languages: C#, TypeScript/JavaScript, SQL (PostgreSQL, SQL Server), Python, Java

Frameworks: .NET, React, Next.js, Node.js, Entity Framework, React Native

Technologies: Terraform, AWS (ECS, ECR, SQS, SNS, RDS, Parameter Store), GCP, Kafka, GraphQL, MassTransit, Docker, Datadog, Dynatrace